

What people actually need from a grocery app

Place order

A usability study of Instacart

CS663 — Human Factors & Usability Metrics

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Everyone just wants to feed their family

Grocery shopping is butter bread for a recipe your grandmother taught you. It's making sure the chicken is halal. It's stretching \$40 to last until Friday.



We sat with 8 shoppers.

Every one of them was trying to do something different.

Home Cook

Feeding family on a budget

Nutrition Focused

Checks calories before price

Vegetarian

Weekly shopper saving time

Halal Shopper

Needs certification to trust

Brand Loyal

Knows her brands, one store

Specialty Hunter

Late-night delivery user

Deal-Driven

Chasing loyalty points

Quality First

Halal, price second

8 Different Lives. 1 Question.

Why did people with eight completely different needs all shop on Instacart in exactly the same way?



How users make product decisions

01

How users evaluate products when shopping online

Price, trust, brand, quality

02

How users choose which store to shop from

Familiarity, fees, distance, reputation

03

How the interface supports or blocks decision-making

Navigation, filters, product clarity

How We Listened

8 Moderated Sessions

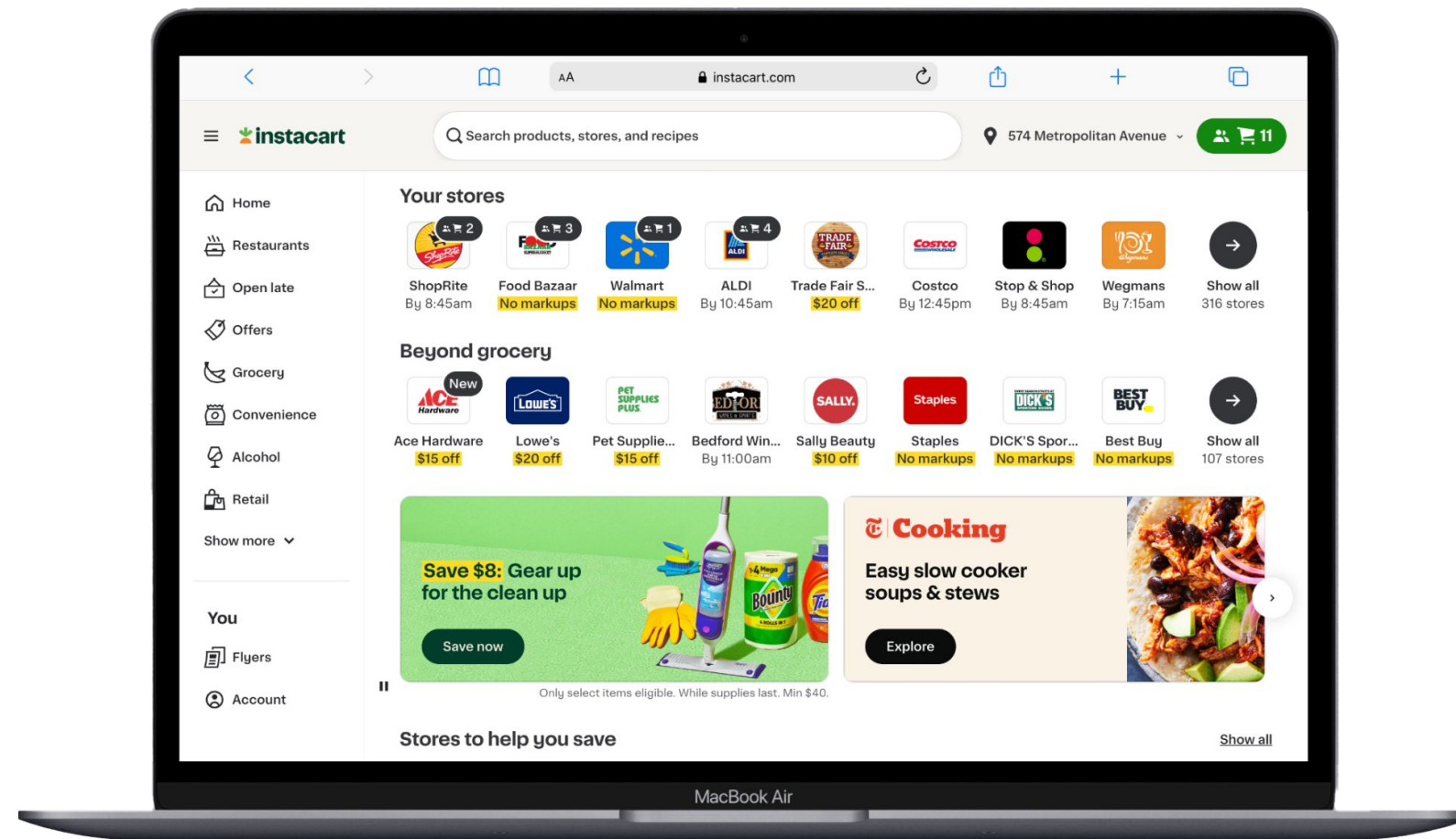
40–50 minutes each

5 Novel + 3 Experienced

Users with varying familiarity

3 Tasks

Last one: no search



Here is what we found...

Familiarity & Routine

Users default to familiarity — in stores and in products. Trust, not comparison, drives choice.

8/8 users chose a store they already knew — before the page finished loading

Loyalty programs and past purchase history reinforced store lock-in

Product choices mirrored in-store habits — familiar brands were trusted by default

I want my points at No Frills!

— novel user

If it's cereal or spices, yeah, I'd try a deal. But for salmon or shrimp — no. I'm getting it from a place I feel comfortable.

— novel user

Search Is a Hidden Dependency

The dominant interaction path for product discovery — and a single point of failure.

When search was available

Tasks 1 & 2 rated 1–2/5 difficulty. Users moved fast, felt confident. Problems stayed hidden.

When search was removed

Task 3: slower completion, verbal frustration, false negatives. Every browsing weakness surfaced at once.

*"I'm lost, because the
quickest way for me to
search that information... I
need the search bar."*
— experienced user

Consequence

Users are over relying on search, making the other navigation unusable.

Participants couldn't find what they were looking for

Categories don't match user mental models.

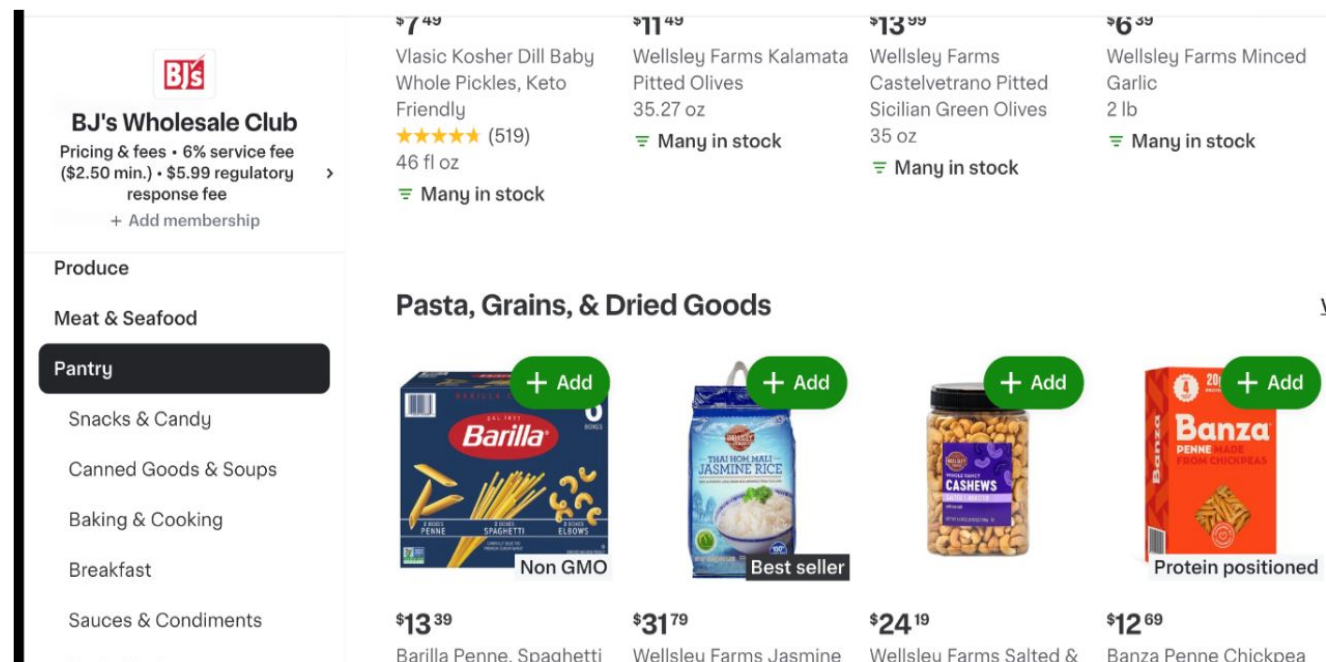
"I don't even know what that would fall under."

— novel user

"Pantry"

Location, not product type

→ **Dry Goods & Pasta**



The real-world store is the mental model.

Users navigate by aisle metaphor — pasta has its own aisle in every supermarket they've ever walked. "Pantry" means a cupboard at home, not a product category.

Filters Failed Users

0 users found
a price filter

3 gave up on
dietary filters

1 store had
gluten-free
filter

MISSING

Dietary filters (gluten-free, dairy-free, low-sodium) absent from most stores

INCONSISTENT

"Grain Free" vs "Gluten Free" — inconsistent labels confused and eroded trust

HIDDEN

"Shop by Preference" section buried in sidebar — users never found it

*Not gonna lie, losing
a little bit of hope.*”
— novel user

Product Cards Don't Surface What People Actually Need

Why don't they have a label for low sodium?

— experienced user

If users don't click into— and most don't — the card IS the product page.

✗ Dietary badges (halal, gluten-free, organic)

✗ Nutritional highlights (calories, sodium)

✗ Certification info (halal, kosher)

✗ Quantity cues matched to usage



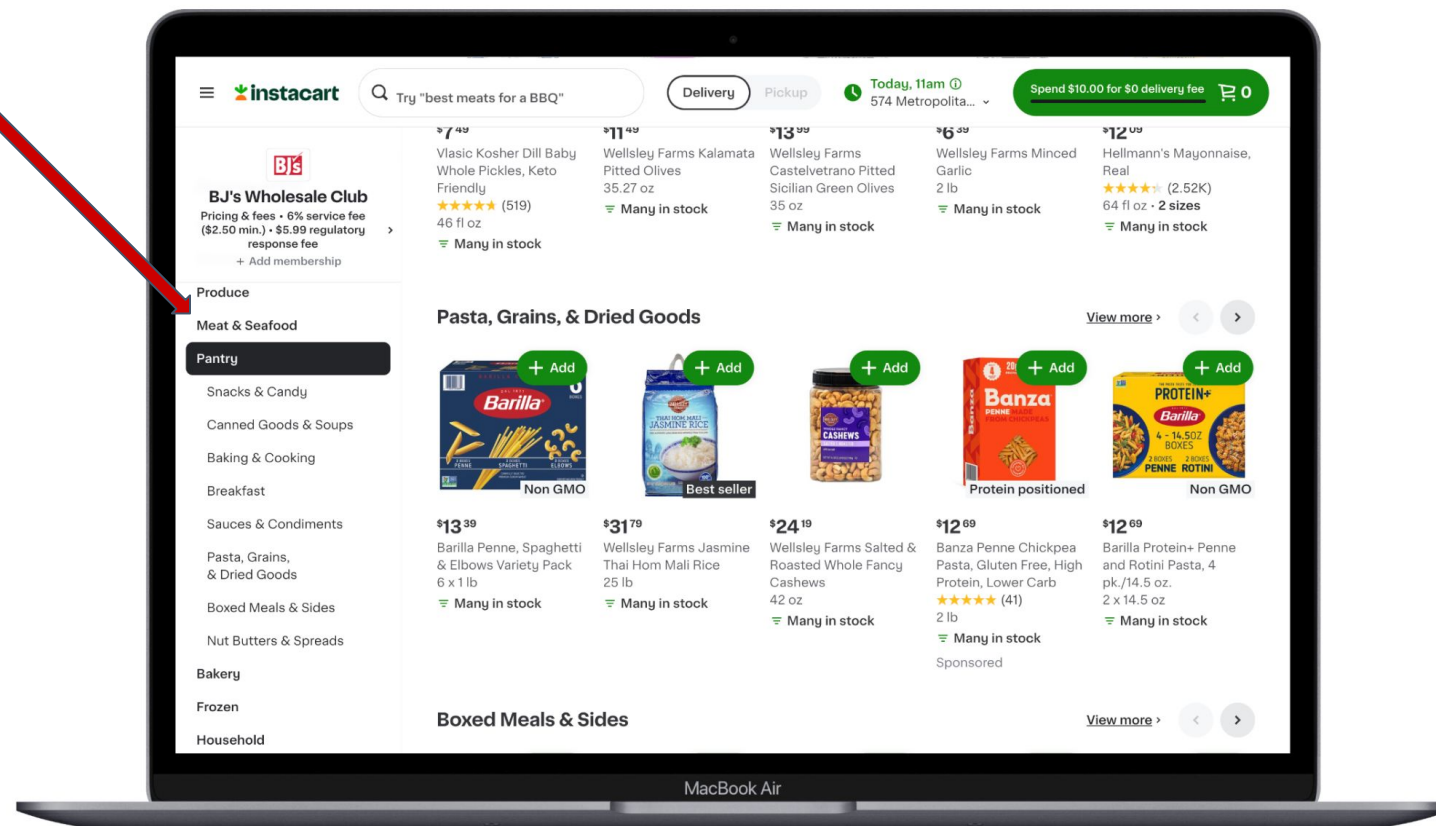


What we saw + Our proposed design solutions

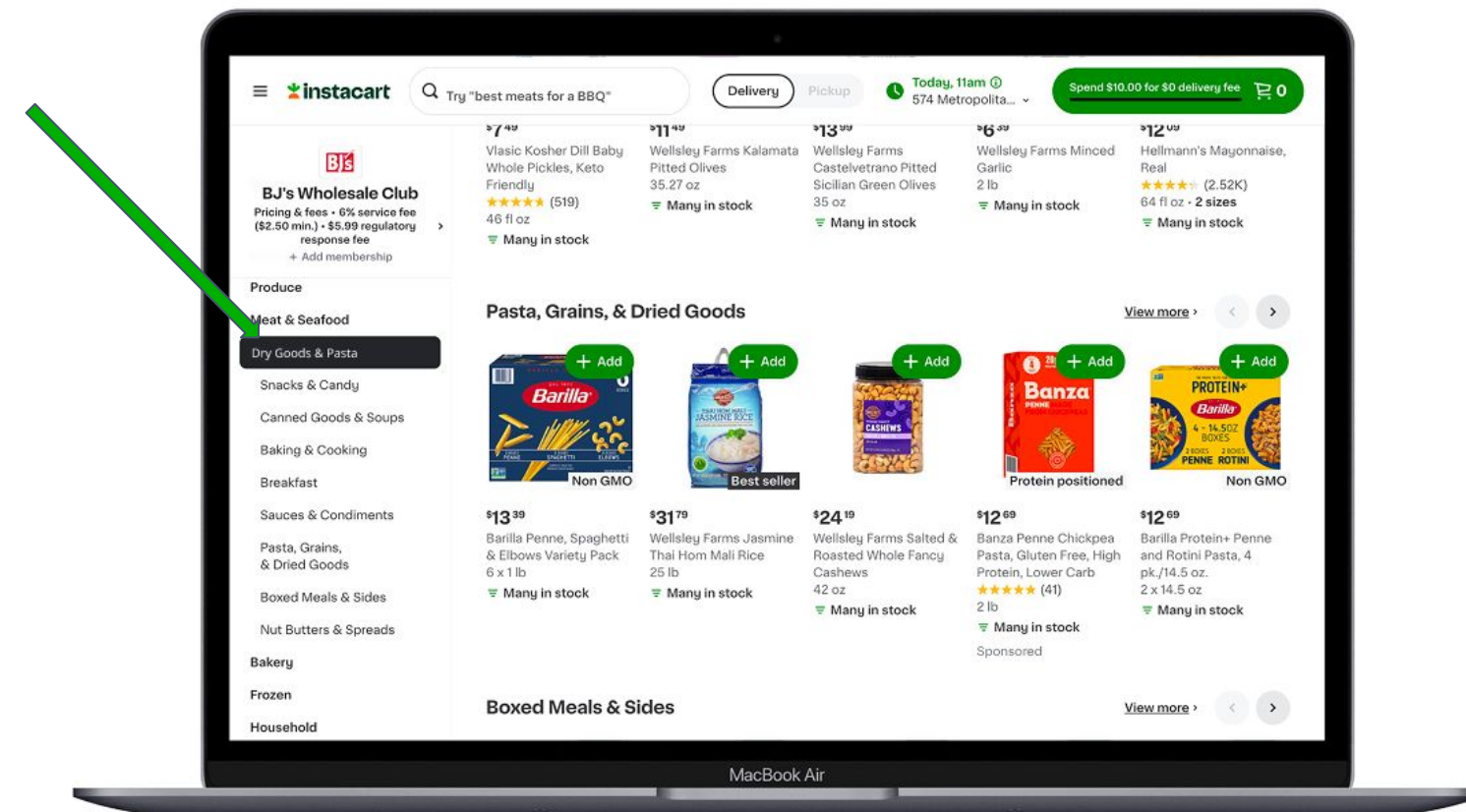
Through the lens of Traps & Tenets

Categories: Align Categories to Mental Models

Trap: Poor Grouping + Inviting Dead End



Current design

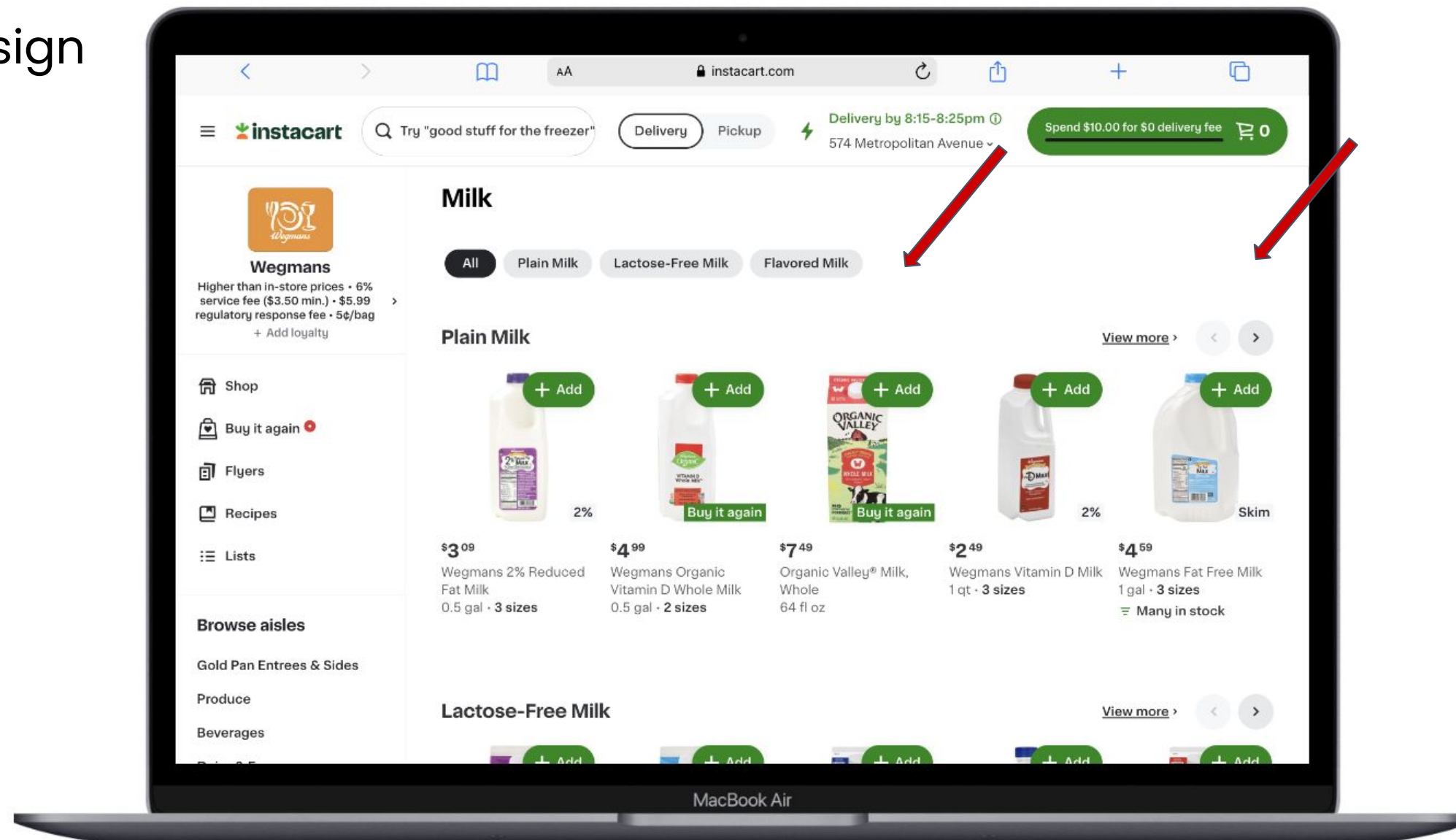


Suggested design

Filters: Consistency and Visibility

Trap: Effectively invisible element:

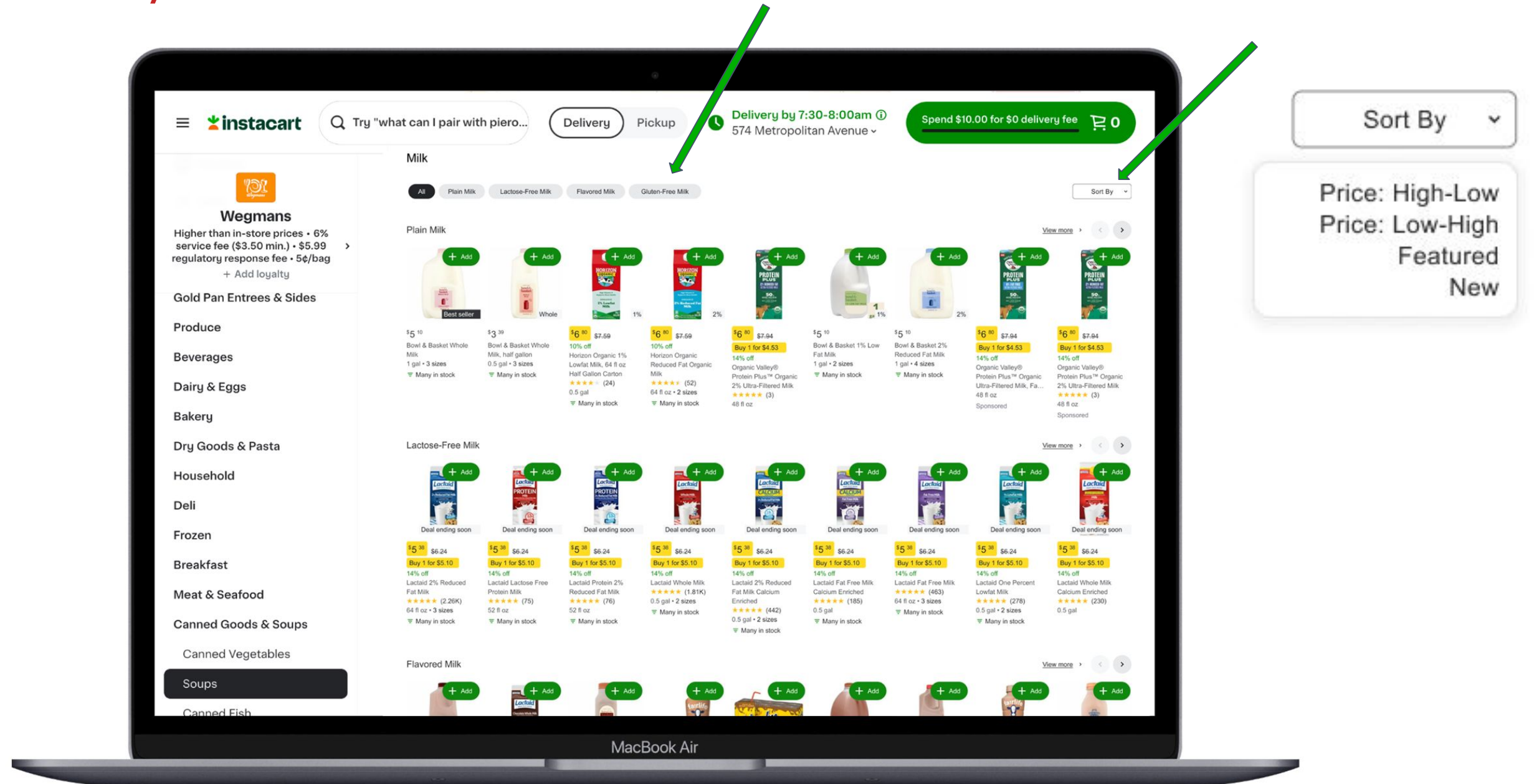
Current design



Filters: Consistency and Visibility

Trap: Effectively invisible element

Suggested design



Product carts: Surfacing Important information



Trap: Effectively Invisible Element + Unnecessary Step



\$5¹⁹

Progresso Reduced Sodium Savory
Chicken & Wild Rice Soup
18.5 oz

☰ Many in stock

Current design



\$5¹⁰

Progresso Reduced Sodium Southwest
Style Black Bean & Vegetable Soup
18.5 oz • 2 flavors

☰ Many in stock

Suggested design

What Instacart reveals about us all

BEHAVIOUR

We shop from memory.

Familiarity isn't laziness — it's cognitive efficiency. We trust what we know because trust is earned by experience, not by interface design. Any system that ignores this fights human nature.

TRUST

Trust breaks quietly.

A mismatched category. A missing badge. A filter that works in one store but not another. Users don't complain — they stop trying. Every friction point is a silent exit.

EQUITY

Not everyone shops the same way.

When a platform can't verify halal certification, or has no results for Ghanaian bread, it isn't just inconvenient. It says: this wasn't built for you. That has real consequences.



Design for the person, not the flow

Every friction point we found has a face behind it.
A shopper who gave up. A parent who felt deceived.
A person whose dietary needs went unseen.

Thank you!